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The Development of a Flow-Through Mode Cation Exchange Process for the Purification of a Monoclonal Antibody

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Abstract

Cation exchange chromatography is typically utilized in bind-and-elute mode for monoclonal antibody purification. However, during purification process development for a novel monoclonal antibody (MAB) intended for clinical use, it was determined that bind-and-elute conditions were not sufficient for removing significant levels of antibody aggregate. Based on preliminary purification data, an alternative purification method, operation of the cation exchange process in flow-through mode, was investigated. Flow-through mode cation exchange conditions were optimized by design of experiments. Optimal conditions resulted in <1% dimer and >85% recovery at all scales evaluated. An additional anion exchange polishing step was required to remove residual process impurities. The possibility of using the flow-through mode, cation exchange approach as a platform process, as well as the benefits and drawbacks of operating the cation exchange process in flow-through mode for MAB purification are discussed.

Introduction

Monoclonal antibodies (MAbs) make up some of the most promising therapeutics for a range of conditions including cancers, autoimmune disorders, and infectious diseases.^[1,2] As of 2013, there were about 350 MAB-based therapies in various stages of clinical trials, and more than 25 MAbs on the market.^[3] The clinical success of MAB-based immunotherapeutics has resulted in significantly growing numbers of products currently in pre-clinical and clinical development for a variety of indications.

In order to minimize process development timelines and speed up time-to-clinic, MAB manufacturers have moved toward a platform approach to purification processes in which a standard set of resins and buffers are used whenever possible.^[4] For the majority of MAbs being developed,

platform processes result in high quality MAB preparations with acceptable levels of product- and process-related impurities such as aggregates and residual host cell proteins (HCP) or DNA.^[5] The standard MAB purification platform includes a protein A capture step and either one or two polishing steps. Two additional safety steps, low pH treatment and nanofiltration, are included for clearing potential viral contaminants.^[5] The most commonly used resins for polishing steps include ion exchange, hydrophobic interaction, and mixed-mode (dual-mode ion exchange, or ion exchange/hydrophobic interaction).^[6]

Cation exchange (CEX) polishing steps are typically operated in bind-and-elute mode where MAB is bound to the resin under low salt conditions and then eluted with a higher concentration of salt.^[7] Unbound and loosely bound impurities either flow through the column during the load step or are removed with subsequent washes while other impurities, specifically aggregates, remain bound to the resin. Conversely, anion exchange (AEX) chromatography for MAB purification is typically run in flow-through (FT) mode^[8,9] in which pH and conductivity conditions are optimized so that only impurities are retained by the resin. The MAB does not bind to the column in the FT mode process. FT mode operation offers several advantages over bind-and-elute mode. Because FT mode processes are designed to bind impurities rather than product, resin loading capacities for the MAB are significantly higher, with throughputs reaching >250 mg/mL resin, while bind-and-elute mode capacities are on the order of 50–100 mg/mL resin.^[9]

A fully human IgG1 MAB that neutralizes diphtheria toxin was isolated from antibody-secreting cells from a volunteer immunized with a diphtheria toxoid (Td) vaccine.^[10] Solubility issues, as a result of self-association, required restoration of the variable region framework sequences for heavy and light chain to the germline consensus sequence.^[11] C4-G antibody produced in CHO cells was purified via a standard platform process with MAB capture on protein A followed by AEX chromatography in FT mode, and then CEX chromatography in bind-and-elute mode. Residual process-related impurities were reduced to

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acceptable levels. However, the levels of product-related impurities, namely aggregate species^[12], remained high after the three-step purification process. During attempts to optimize the CEX process in bind-and-elute mode, it was determined that aggregates were only effectively removed (to <1%) using low salt buffers for elution of the MAb. However, these conditions were not practical for operation as they resulted in low MAb concentrations and large elution pool volumes of >10 column volumes (CVs).

The elution results did, however, indicate that the CEX step could potentially provide adequate aggregate removal if operated in FT mode. The development and optimization of a FT mode CEX process was performed

in two stages. Initially, experiments were run to gain process understanding, then conditions were further refined using high-throughput methods and design of experiments (DOE). The process was then scaled 50-fold to confirm the small-scale results. Using optimized conditions, MAb aggregate levels were reduced to <1% with no negative impact on final residual impurities such as DNA, HCP, and protein A levels. The final FT mode CEX process has several advantages over bind-and-elute mode including reduced: (1) resin volume requirements; (2) processing time; (3) buffer consumption; and (4) process complexity; all of which ultimately reduce the overall cost of producing a MAb.

Materials and Methods

Materials

All MAbs used in this study were fully human IgG1 MAbs that were either derived from human antibody-secreting cells or HuMAb mice. The heavy and light chain constant regions for all four MAbs are the IgG1 germline sequence.

The MAbs were produced in Chinese hamster ovary (CHO) cells. The cells were clarified by depth filtration through [Millistak+® D0HC](#) and [A1HC Pod](#) filters (EMD Millipore) and the cell culture supernatant was purified over a [ProSep®-vA Ultra](#) (EMD Millipore) chromatography column. For the protein A purification, the column was loaded and washed at neutral pH, and the protein was eluted with a low pH buffer. The protein A pool was then adjusted to pH 5.0 with 2 M Tris base and 0.2 µm filtered prior to storage at 4°C. A280, pH, and conductivity were continuously monitored during chromatography runs.

All CEX purifications were performed on [Toyopearl GigaCap S-650M resin](#) (Tosoh Bioscience LLC). The GigaCap S-650M resin was packed in [Benchmark chromatography columns](#) (Omnifit®) of varying diameters depending on the scale. All buffers and chemicals used were [J.T. Baker® products](#) (Avantor™ Performance Materials). AEX purification was performed on [Mustang Q](#) (Pall Corporation) membranes. All purification runs were performed on [AKTApurifier 100](#) (GE Life Sciences) or [BioRad BioLogic DuoFlow 40](#) (Bio-Rad Laboratories) fast protein liquid chromatography (FPLC) systems. Chromatography parameters are described in the methods section.

Methods

FT Mode CEX Purification Process of C4-G MAb

The FT mode CEX process consists of the following steps: (1) charge with 1 M NaCl; (2) equilibrate with appropriate buffer; (3) load protein; (4) wash with equilibration buffer until UV absorbance returns to baseline; and (5) clean-in-place (CIP) with 1 M NaCl. The MAb was adjusted to the same

pH and conductivity as the equilibration/wash buffer for each experiment. With the exception of the load and wash steps, all steps were performed in accordance with the bind-and-elute mode experiments unless otherwise specified.

DOE for FT Mode CEX Optimization

A 16-run response surface model DOE was created using [JMP® 9 statistical software](#) (SAS Institute, Inc.). The conditions for each run are listed in **Table 1**. The MAb was analyzed for percent of recovery and percent of dimer reduction by size-exclusion, high-pressure liquid chromatography (SEC-HPLC). Optimal conditions for the FT mode CEX process were determined using the “maximize desirability” function in JMP. The safe operating window was determined using the JMP contour profiler.

TABLE 1. Conditions for the optimization of the FT mode CEX process by DOE.

Run #	pH	NaCl (mM)	Load Flow Rate (cm/h)	Load Amount (mg/mL resin)
1	7.0	50	87	375
2	7.0	75	125	375
3	6.0	75	87	500
4	7.0	100	87	500
5	6.0	100	87	375
6	6.0	100	200	500
7	5.0	50	87	250
8	5.0	100	125	250
9	6.0	50	200	250
10	6.0	75	125	375
11	7.0	50	200	500
12	7.0	100	200	250
13	5.0	100	87	500
14	5.0	75	200	375
15	5.0	50	125	500
16	7.0	75	87	250

Scale-Up of FT Mode CEX Process

Protein A-purified C4-G was loaded onto a 51.5 mL column packed with Toyopearl GigaCap S-650M resin. Prior to loading, the MAb was adjusted to a conductivity of 11.0 mS/cm with 1 M NaCl, pH 5.75, and 2 M Tris base. The resin was equilibrated and washed with 20 mM sodium citrate, pH 5.75, and 80 mM NaCl. The MAb was loaded to 350 mg/mL resin and a flow rate of 125 cm/h. The flow-through and wash was fractionated into 200 mL fractions until the A280 reached baseline. The flow-through and wash fractions were analyzed for recovery and purity by SEC-HPLC.

AEX Membrane Purification with Mustang Q

The CEX-purified C4-G, purified by protein A, and followed by CEX, was adjusted to pH 7.5 with 2 M Tris base, and the conductivity was adjusted to 3 mS/cm by dilution with ultrapure water from the [Milli-Q® water purification system](#) (EMD Millipore). The MAb was loaded onto a 10 mL Mustang Q membrane. The membrane was charged with 1 M sodium chloride (120 mL, 75 mL/min) and equilibrated with 120 mL of 20 mM Tris, pH 7.5, until the pH and conductivity equilibrated. The membrane was loaded with 14 g of MAb and the membrane was washed with 20 mM Tris, pH 7.5, until the UV reading reached baseline.

Purification of Three Additional MABs by FT Mode CEX Process

The optimal conditions for the purification of the three additional MABs tested were determined from a buffer screen in 96-well plate format (1 mL [AcroPrep™ 0.45 µm filter plates](#), Pall Corporation). MABs B, C, and D are also fully human IgG1 antibodies. The plates were prepared such that each well contained 15 µL of GigaCap S-650M resin. The equilibration/wash buffers were all prepared in 20 mM sodium citrate and ranged from pH 5.0–6.5 with 0–125 mM NaCl. The buffer conditions that were chosen to move forward were decided upon based on MAB recovery and purity by SEC-HPLC.

The optimal conditions determined from the high-throughput screen were scaled up to a 16.6 mL column for MABs B, C, and D. All conditions were consistent with those used for C4-G (MAB A) except for the conductivity and pH of the MAB, equilibration, and wash buffers. The equilibration buffer used for the purification of MABs B and D was 20 mM citrate, pH 6.0, 10 mM NaCl, and each MAB was adjusted to 5–6 mS/cm with ultrapure water. MAB C was purified using 20 mM citrate, pH 6.5, as the equilibration and wash buffers, and the MAB conductivity adjusted to 4 mS/cm with ultrapure water.



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MAB Purity and Recovery by SEC-HPLC

MAB species (dimer, shoulder, and monomer) were quantified by SEC-HPLC using a [TSKgel G3000SWxl column](#) (Tosoh Bioscience LLC). An isocratic elution (1.0 mL/min) was performed in phosphate-buffered saline (PBS), pH 6.5, with 0.02% sodium azide on a Waters [1525 Binary HPLC Pump](#). Protein peaks were detected at 280 nm using a Waters [2487 dual wavelength detector](#). Protein

concentration was estimated based on standard curves generated for each of the MABs studied.

Assays for Residual Impurities: CHO Host Cell Protein, DNA, and Protein A

Commercially available kits were used according to manufacturers' protocols to analyze residual HCP, DNA, and protein A.

Results and Discussion

Initial Evaluation of FT Mode CEX Process

A full resin screen for reducing C4-G aggregate levels was not performed due to time restrictions for the project. The development of a FT mode CEX process was an obvious next step once it was realized that bind-and-elute mode operation would only result in acceptable dimer removal at low salt concentrations. Using conditions derived from an elution gradient, a proof-of-concept run was performed in which the GigaCap S-650M resin, equilibrated with 20 mM citrate, pH 6.0, and 75 mM NaCl, was loaded to 80 mg/mL resin with C4-G. The flow-through and wash

was fractionated in order to determine collection cutoffs to maximize recovery and purity.

The initial FT mode CEX run (**Figure 1**) resulted in a typical FT mode chromatogram with the MAB eluting in the flow-through, and the wash fractions and the majority of the dimer and aggregates eluting with the high-salt CIP step. **Table 2** shows the SEC-HPLC results for the flow-through and wash fractions combined, the tail of the wash fraction and the CIP. The dimer levels in the flow-through/wash fraction were reduced from 2.43% to 1.06% with no optimization. Also notable is that the CIP peak was enriched

in dimer (56.5%), confirming that the dimer binds tighter to the CEX resin than monomeric MAB.

Unlike the FT mode AEX process for MAB purification where the MAB does not bind to the resin and impurities bind tightly, C4-G does interact slightly with the column under the conditions used for FT mode CEX operation. As a result, an extended wash step was needed. The aggregates bound tighter to the CEX resin than monomeric MAB but eventually eluted later in the wash step (see "wash tail" in **Table 2**), adding to the separation complexity. In order to achieve maximal recovery and purity, a strategy for peak collection based on A280 cutoffs was developed. Following an SEC-HPLC purity analysis of flow-through and wash fractions from several experiments, it was determined that the MAB should only be collected until a particular UV cutoff was reached to ensure that the MAB pool was >99% monomer. Peak cutoffs were only determined to maximize MAB monomer purity. Process-related impurities were not considered at this point in development.

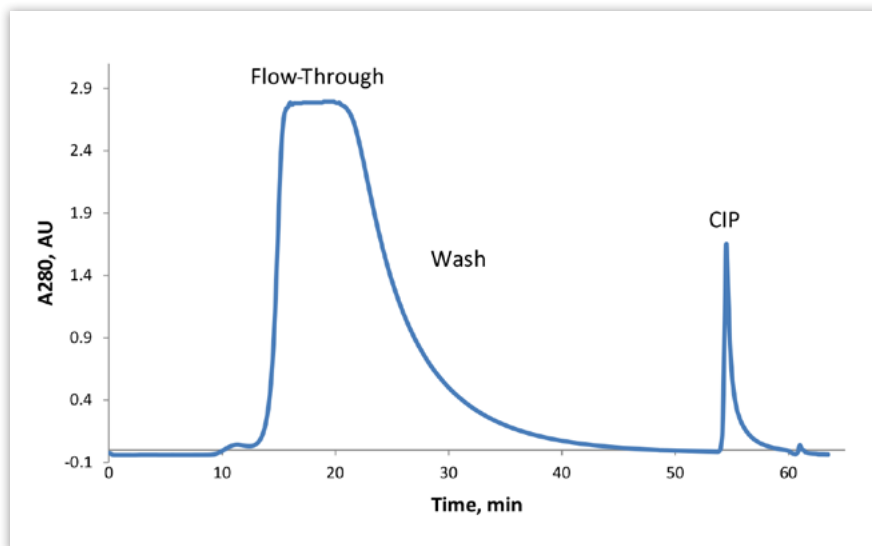


FIGURE 1. FPLC chromatogram (280 nm) for the initial FT mode CEX purification run. The flow-through, wash, and CIP peaks are labeled.

TABLE 2. SEC-HPLC results for initial FT mode CEX run.

Sample	Polymer (%)	Dimer (%)	Monomer (%)	Recovery (%)
Load	0.68	2.43	96.88	—
Flow-Through/Wash	—	1.06	98.94	83.0
Wash Tail	0.21	2.90	96.89	1.6
CIP	8.72	56.49	34.79	—

Process Optimization with DOE

In order to optimize the FT mode CEX process for dimer reduction, and to define an acceptable working space, a 16-run response surface model DOE was initiated. Based on process knowledge gained in the previous experiments, the factors investigated were: (1) load/equilibration pH (5.0–7.0); (2) load/equilibration NaCl concentration (50–100 mM); (3) load amount (250–500 mg/mL resin); and (4) load flow rate (87–125 mL/min). In each case, the MAb was adjusted to match the pH and conductivity conditions of the equilibration buffer. All runs were performed on a 1.1 mL GigaCap S-650M column. The FT and wash were collected using an A280 cutoff that was determined during initial screening experiments. The results were analyzed for recovery and MAb purity (dimer reduction) by SEC-HPLC.

The significant main effects for achieving maximal MAb recovery were load pH, NaCl concentration, and load amount. For maximal dimer reduction, load pH and flow

rate were the most important. Interestingly, no interaction terms were significant from this analysis. Pareto plots for MAb recovery and dimer reduction are shown in **Figure 2**. Pareto plots represent the overall impact of a particular input factor on the response (recovery or dimer reduction). Load pH has the greatest impact (~40%) on both MAb recovery and dimer reduction.

Using the “maximize desirability” function in JMP9, the optimal conditions for the FT mode CEX process were determined to be an equilibration/load buffer of 20 mM citrate, pH 5.75, 80 mM NaCl, with a load density of 500 mg MAb/mL resin loaded at a flow rate of 125 cm/h. The predicted recovery and dimer percentages under these conditions were 83% and 1.15% respectively. The optimal conditions were tested on the same 1.1 mL column and resulted in 84% recovery with 1% dimer, confirming the reliability of the DOE model.

According to the DOE analysis, the acceptable working ranges needed to achieve >80% recovery and >50% dimer

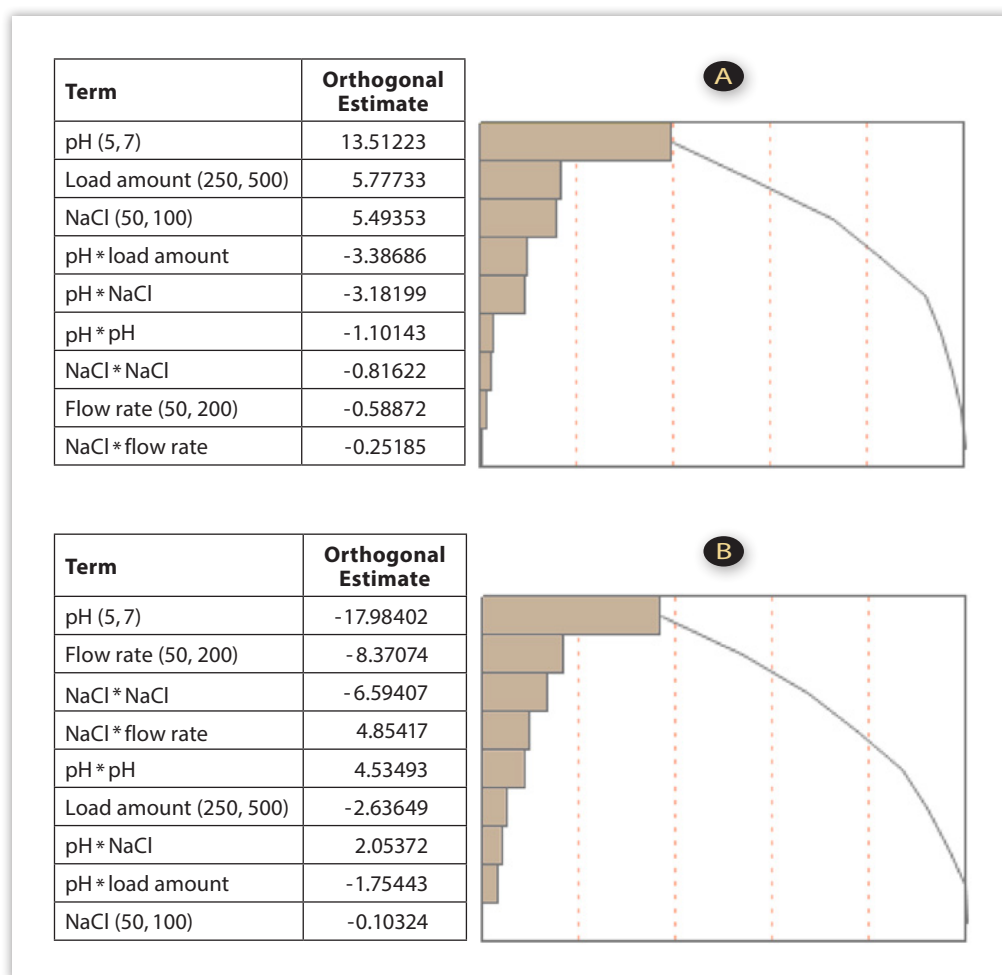


FIGURE 2. Pareto plots from the FT mode CEX DOE performed, ranking the factors (from top to bottom) with: **(A)** the greatest influence on recovery; and **(B)** dimer reduction.

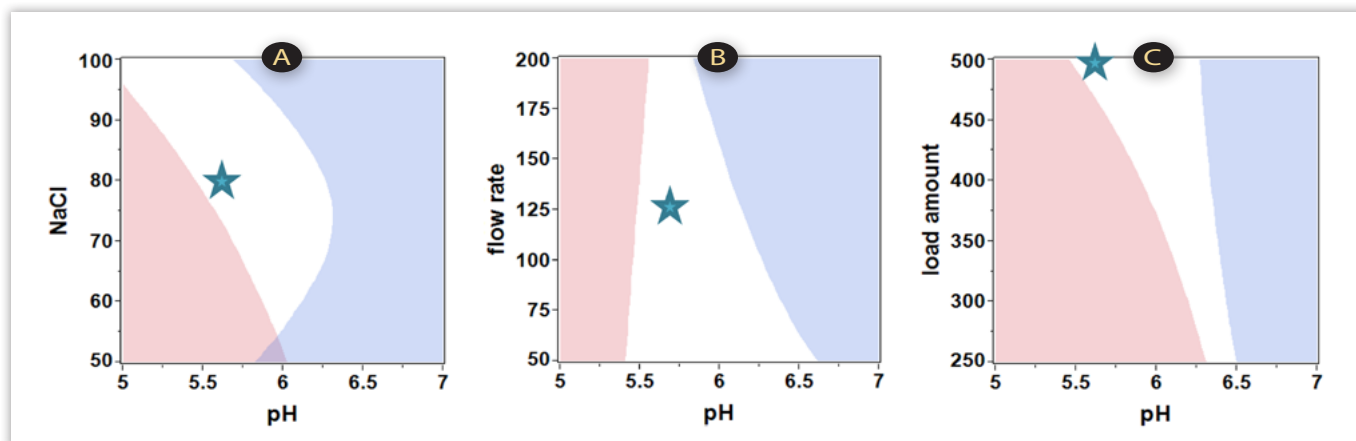


FIGURE 3. Contour plots generated from the JMP9 contour profiler for: **(A)** NaCl; **(B)** flow rate; and **(C)** load amount. The red and blue shaded areas represent conditions that would result in <80% recovery and <50% dimer reduction, respectively. The blue stars designate the optimal conditions determined by the JMP9 analysis.

reduction (**Figure 3**, white areas) are: (1) pHs from 5.75–6.25; (2) NaCl from 75–85 mM; (3) flow rates from 50–125 cm/h; and (4) load amounts from 400–500 mg/mL. Five runs using conditions covering the acceptable working range were completed. All results were consistent with the model's predicted results (data not shown).

Scale-Up of FT Mode CEX Process

The FT mode CEX process was scaled 50-fold to accommodate C4-G antibody produced in a 10 L bioreactor. The chromatogram is shown in **Figure 4**. The flow-through and wash steps were fractionated to ensure that the MAb purity in the final pool was acceptable. Ultimately, all flow-through and wash fractions were combined to yield 95.4% recovery with 99.1% monomer and 0.9% dimer.

Residual Process Impurities

When switching chromatography unit operations, an obvious concern is the impact on the residual process impurities. The levels of residual HCP, DNA, and protein A after bind-and-elute and FT modes in a CEX operation were determined (**Table 3**). As expected, DNA levels were significantly reduced in the bind-and-elute mode because DNA was removed during load and

wash steps. HCP levels were not reduced to an acceptable level using either mode of operation, indicating that an additional polishing step following CEX is necessary, regardless of the CEX process used. Protein A levels were reduced significantly in FT mode, but only slightly in bind-and-elute mode. This was expected since leached protein A binds tighter to CEX resin than C4-G. Therefore, in FT mode, there is less residual protein A associated with the MAb as compared to bind-and-elute mode in which the HCP elutes with the MAb.^[7]

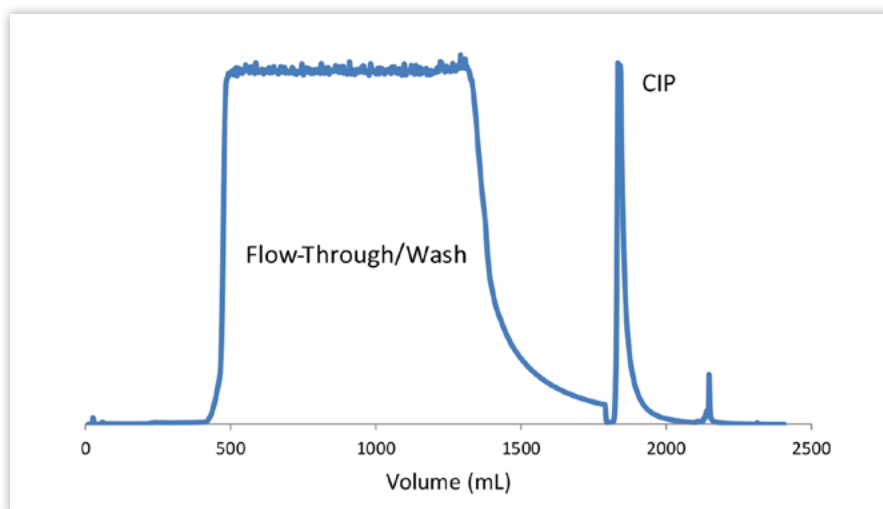


FIGURE 4. FPLC chromatogram (A280 trace) for scale-up FT mode CEX purification. The flow-through/wash and CIP peaks are labeled.

TABLE 3. Residual C4-G antibody process impurities* measured to facilitate optimization of the CEX process in bind-and-elute and FT modes.

C4-G Impurities	C4-G Bind-and-Elute Mode	C4-G FT Mode
HCP	28.3% (210 ppm)	17.0% (279 ppm)
DNA	97.6% (0.5 ppb)	29.2% (15 ppb)
Protein A	11.7% (53 ppm)	97.6% (1.5 ppm)

*Actual values shown in parentheses.

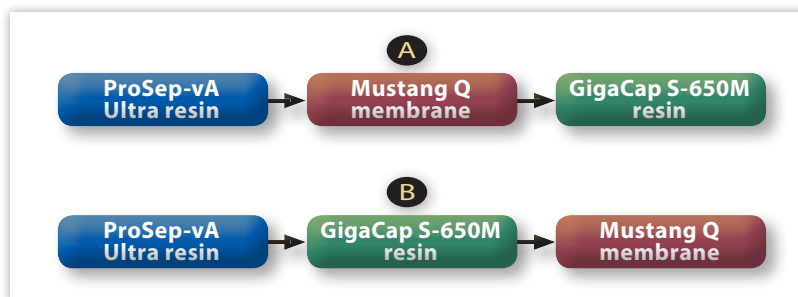


FIGURE 5. (A) Original MAb purification platform used for bind-and-elute mode CEX. **(B)** Modified process to accommodate FT mode CEX.

Incorporation of FT Mode CEX into a Three-Step Purification Process

Process efficiency is more important than ever, now that downstream processing is currently considered to be the bottleneck in the production of biologics.^[4] It has, therefore, become a priority to streamline downstream purification processes, as evidenced by the push for continuous manufacturing operations.^[13,14] Because the conductivity of the CEX equilibration buffer was close to the conductivity of the MAb following protein A chromatography and low pH treatment (**Figure 5**), the order of the CEX and AEX steps was switched from the original platform process. This simple change maximizes process continuity and minimizes the manipulation of the MAb pool prior to loading the CEX column. The three-step purification process was tested using the following: ProSep-vA Ultra resin → GigaCap S-650M resin → Mustang Q membrane. The results of this purification process met all specifications. An overall recovery of 78.5% was achieved with 99.1% monomer, 9.0 ppm HCP, and <1.6 ppb DNA.

FT Mode CEX as a Platform Process

Given the time and cost-saving benefits of operating chromatography processes in FT mode, this process was evaluated for three additional MABs. FT mode CEX conditions were first optimized in high-throughput format using 96-well plates containing GigaCap S-650M resin. Optimal conditions were then scaled up to 1 mL columns for further evaluation. As can be seen in **Table 4**, the FT mode CEX

process was capable of reducing dimer levels by >50% for MABs B and C. Based on these results, the efficiency of FT mode operation appears to be limited by the type of dimer present and the amount of dimer in the load.^[15]

We believe that there are different modes of self-association in C4-G compared to MABs C and D. The C4-G MAB has an unusually high pI of 9.6 and is the least hydrophobic of the MABs used in this study^[11], while MABs C and D are the most hydrophobic of the four MABs. Since C4-G has a high pI and low hydrophobicity, the aggregate formed must largely be due to electrostatic interactions.^[16] It then follows that self-association could be driven by electrostatics in C4-G and hydrophobicity in MABs C and D. This would explain the difference in the extent of dimer removal following the FT mode CEX process for the four MABs used in this study. Additional MABs will be evaluated to determine if there is a relationship between MAB hydrophobicity and efficiency of dimer removal by FT mode CEX. If there is a link, there is the possibility of using sequence information to predict whether a MAB is a good candidate for FT mode CEX.

Residual Impurities After Three-Step Purification Process

Residual protein A, HCP, and DNA levels were determined for C4-G and MABs B, C, and D (**Table 3**). In general, the impurities were reduced to acceptable levels, or levels that can be further reduced to fall within acceptable levels by adding a polishing step, as was observed for C4-G.

Impurities	C4-G	MAB B	MAB C	MAB D
Dimer	57.0% (0.9%)	52.0% (0.9%)	87.0% (0.4%)	70.0% (0.3%)
HCP	17.0% (279 ppm)	74.3% (36 ppm)	81.3% (25 ppm)	20.5% (184 ppm)
DNA	29.2% (15 ppb)	90.4% (15 ppb)	29.1% (6.1 ppb)	90.0% (10 ppb)
Protein A	97.6% (1.5 ppm)	0% (51 ppm)	92.9% (1.0 ppm)	15.3% (3.1 ppm)

*Actual values shown in parentheses.

Conclusions

An effective and efficient FT mode CEX process was developed for the purification of a MAb in which dimer levels could not be reduced to acceptable levels with standard bind-and-elute mode operation. While FT mode CEX is a diversion from the standard MAb CEX purification, the same resin and similar buffers are utilized no matter which mode of operation is chosen. The benefits of FT mode operation include a 7-fold greater dynamic binding capacity, at least a 40 % increase in productivity, and 75 % lower amounts of buffer consumption.

There are several drawbacks that should be considered before switching to FT mode CEX, however. The process results in a 6-fold greater MAb pool and the removal of process impurities is less efficient. Also, implementation of the FT mode CEX process is dependent on the properties of the incoming feed stream and the flexibility of adding an ultrafiltration step prior to nanofiltration to reduce the load volume on the filter. Although the FT mode CEX process may not be amenable to all MAbs, having both modes available for use broadens the utility of the platform process.

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